# Review 2 Commitment

## 1. Project Information

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|---|---|
| **Project code** | GFA26SE153 |
| **Project name** | Laboratory Information Management Subsystem for Contract Testing Laboratory on Odoo Platform (Phân hệ Quản lý Thông tin Phòng kiểm nghiệm dịch vụ trên nền tảng Odoo) |
| **Team** | Nguyễn Nam Hoàng SE183190 (leader), Lê Sỹ Tuyền SE190797, Đinh Gia Huy SE193759, Phạm Anh Kiệt SE190051, Lê Nguyễn Hoàng Anh SE184416 |
| **Supervisor** | Nguyễn Thế Hoàng (hoangnt2@fpt.edu.vn) |
| **Review 2 date** | 24/10/2026 |
| **Reviewer 1** | To be completed at Review 2 |
| **Reviewer 2** | To be completed at Review 2 |
| **Version** | 1.1 |
| **Date** | 09/11/2026 |

## 2. Final Registration Form

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|---|---|
| **Final version** | 2.0 (per Form BM03 – Record of Changes) |
| **Approved at Review 2 (Yes/No)** | Pending Review 2 confirmation |
| **Attached file name** | SWP490_GFA26SE153_R2_RecordOfChanges.docx |

## 3. Mandatory Features

| ID | Feature | Actor | Acceptance criteria | Workflow | Owner |
|---|---|---|---|---|---|
| F-04 | Portal cart, drafts and reorder | Customer | Main: A visitor fills a cart without signing in and submits it with their details. Rule: A package is ordered whole; reordering creates a draft at current prices and leaves the original unchanged. Exception: Submitting without an e-mail address is refused. | WF-01 | Lê Sỹ Tuyền |
| F-05 | Order claiming and orders on behalf | Sales | Main: A salesperson claims a submitted order, or creates one for a customer who called. Rule: A salesperson sees only their orders and the unclaimed queue; the Head of Sales sees all. Exception: An order already claimed cannot be claimed by another salesperson. | WF-01 | Lê Sỹ Tuyền |
| F-06 | Quotation and revisions | Sales | Main: Sales confirms the ways and the expected samples calculated from the sampling requirements, applies a percentage discount and sends the quotation. Rule: Each item copies its price, discount and tax rate, so a sent quotation never changes; a revision makes the old version read-only. Exception: A quotation without tests cannot be sent, nor one above the Head of Sales's thresholds before it is approved. | WF-01 | Lê Sỹ Tuyền |
| F-09 | Advance invoice and online payment | Customer | Main: Sending a quotation with an advance creates exactly one posted advance invoice; paying it accepts the quotation. Rule: Online payment accepts any amount up to the remaining balance; a bank transfer or cash payment counts once the accountant confirms it. Exception: A repeated PayOS notification records no second payment. | WF-01 | Lê Sỹ Tuyền |
| F-10 | Quotation approval and order confirmation | Sales, Head of Sales | Main: A quotation above the Head of Sales's thresholds is approved before it is sent; confirming the accepted order creates its test request with the calculated expected samples. Rule: Quotations within the thresholds and confirmed orders need no approval; the customer pays the balance before testing unless they may order on credit. Exception: The salesperson who requested approval cannot approve; a declined quotation goes back to Sales with the reason; a quotation awaiting approval cannot be edited. | WF-01 | Lê Sỹ Tuyền |
| F-15 | Test requests and lab codes | Sample delivery staff | Main: Staff create a test request with its samples, parameters, expected date and report delivery. Rule: Each sample gets a unique lab code, numbered again each year; a request staff create skips the paid step. | WF-02 | Nguyễn Nam Hoàng |

| ID | Feature | Actor | Acceptance criteria | Workflow | Owner |
|---|---|---|---|---|---|
| | | | Exception: Confirming without a sample, or a sample without a parameter, is refused; testers see only the lab code. | | |
| F-16 | Sample collection and handover record | Sample delivery staff, Customer | Main: Staff claim the collection task and plan its time; it appears on their schedule and to-do list, with a reminder. Rule: The customer confirms the actual samples and signs the handover record before labels are printed; signing stores the signature, time and device and e-mails the PDF. Exception: A signed handover record cannot be edited; a correction is a new record. | WF-02 | Nguyễn Nam Hoàng |
| F-17 | Reception, labels and department handover | Sample delivery staff, Head of department | Main: The condition is checked first; a sample in good condition is recorded as received with its time, amount and storage, and one testing task per sample test is queued. Rule: The label shows the lab code, request code, due date and QR code, never the customer. Exception: A label scanned by a person outside the department is refused. | WF-02 | Nguyễn Nam Hoàng |
| F-18 | Insufficient and rejected samples | Sample delivery staff, Head of department | Main: An insufficient sample goes on hold before receipt and its head of department is told. Rule: Asking for more creates a collection task recording who asked and when; more sample returns it to received. Exception: An unusable sample is rejected by the system with its condition note and gets no testing task. | WF-02 | Nguyễn Nam Hoàng |
| F-23 | Task assignment and claiming | Head of department, Tester | Main: Assigning a sample assigns all its testing tasks in the department to one person, who is notified. Rule: An assigned task leaves the queue and appears in the assignee's to-do list. Exception: A claimed task cannot be claimed again; another department's task cannot be claimed; reassigning without a reason is refused. | WF-03 | Phạm Anh Kiệt |
| F-24 | Booking proposal and head approval | Head of department, Tester | Main: The system proposes a booking of the agreed way on the fit machine that finishes soonest before the due date; when it is feasible, the head approves it, the tester is notified, and the tester, required locations and machine are reserved together. Rule: A booking takes the run time rounded up to 15-minute slots; urgent tests first, then first in first out. Exception: Bookings of one machine never overlap; with no feasible booking the head reviews other dates, planned OverTime or outsourcing. | WF-03 | Phạm Anh Kiệt |
| F-25 | Schedules and reminders | Tester, Sample delivery staff | Main: A booking appears on the person's calendar, the machine's schedule and the to-do list. Rule: A reminder arrives 15 minutes before it starts, once on each channel. Exception: An event the administrator turned off notifies nobody; a mandatory event cannot be turned off. | WF-03 | Phạm Anh Kiệt |
| F-30 | Testing method steps | Administrator, Tester | Main: The tester sees the method's steps with number, duration, what to do and note, in number order. Rule: Two steps of one method cannot share a number. Exception: A tester cannot change a step. | WF-04 | Đinh Gia Huy |
| F-32 | Regulations and limits | Administrator, Tester | Main: When a regulation comparison is requested, the result shows the related limit copied with its unit and is assessed. Rule: Later changes to the regulation leave existing results unchanged; a regulation holds one limit per parameter. Exception: A result with a requested comparison cannot be sent without its assessment. | WF-04 | Đinh Gia Huy |
| F-34 | Result entry and approval | Tester, Head of department | Main: The tester enters, signs and sends a result, which closes the testing task; the head approves it. Rule: A rejected result is kept, and a retest adds a new result with a new booking; only the latest is sent for approval. Exception: A tester cannot approve their own result, but the head may approve an entry they made; a rejection opens a new testing task and tells the tester the reason. | WF-04 | Đinh Gia Huy |
| F-35 | Unusual result warning | Tester | Main: A numeric result far from approved results of the same parameter, way of testing | WF-04 | Đinh Gia Huy |

| ID | Feature | Actor | Acceptance criteria | Workflow | Owner |
|---|---|---|---|---|---|
| | | | and sample type warns before sending. Rule: A result near the regulation limit warns; with too few approved results only the limit check runs. Exception: Sending anyway keeps the warning, which the approving head sees; the warning never names a customer. | | |
| F-36 | Sample test cancellation | Head of department | Main: The head cancels a waiting, assigned or testing sample test with a reason; its task and bookings are cancelled. Rule: A sample whose remaining tests are approved completes; a cancelled test is left out of the report. Exception: Cancelling without a reason, or a test with an approved result, is refused. | WF-04 | Đinh Gia Huy |
| F-38 | Outsourcing suggestions and dispatches | Head of department, Sample delivery staff | Main: Accepting an outsourcing suggestion, or outsourcing a test urgently with a reason, puts its sample tests in a dispatch; when staff carry it, the system records the date sent, who took it and the expected date. Rule: Only a subcontracted way with the test's parameter and method can be chosen; the customer is neither told nor asked. Exception: A dispatch whose tests have results cannot be cancelled; a failed delivery goes back to sample reception for the same sample. | WF-05 | Lê Nguyễn Hoàng Anh |
| F-40 | Subcontractor results | Subcontractor | Main: The subcontractor receives the samples and enters the results of the tests sent to it, and the head is notified. Rule: The result shows that user as its author and goes straight to the head for approval like any other. Exception: A subcontractor user sees only its own dispatches and never a sample's customer. | WF-05 | Lê Nguyễn Hoàng Anh |
| F-43 | Report signing and digital signature | Lab head | Main: When every test of the request is approved or cancelled, the system creates the read-only draft report, validates its data, reserves a number and prepares the PDF in the request's language; the lab head reviews it, the PDF is digitally signed and stored, and issue makes the number official. Rule: The report prints the approving heads and the lab head, not the testers, and leaves out cancelled tests. Exception: With Viettel Sign down nothing is marked signed and signing is retried; rejecting without a reason or a named result is refused. | WF-06 | Phạm Anh Kiệt |
| F-45 | Report delivery and download | Sample delivery staff, Customer | Main: Staff record the delivery of an issued report; an e-mail delivery sends the PDF. Rule: An issued report waits in the delivery list until its delivery is recorded. Exception: An unissued report cannot be delivered; a customer downloads only their own reports. | WF-06 | Phạm Anh Kiệt |
| F-47 | Notifications and preferences | Customer | Main: An issued report notifies the customer on the channels they chose. Rule: A new customer is notified by e-mail without configuring anything; a changed preference applies to the next notification. Exception: A mandatory notification cannot be turned off. | WF-06 | Phạm Anh Kiệt |
| F-53 | Laboratory folders and sync | Tester | Main: Saving a file in a folder bound in Medilab Sync adds a version with the tester, the computer, the size and the hash. Rule: Only files matching the folder's sync rules are uploaded; content the file already has is not stored twice. Exception: A person who is not a member of the folder, or cannot see its sample, gets nothing from it. | WF-04 | Đinh Gia Huy |
| F-54 | File versions, conflicts and restore | Tester | Main: Restoring version 3 of a file with 5 versions adds version 6 with the content of version 3. Rule: A file uploaded from an older version is kept as a conflict version and the current version stays. Exception: No one can change or delete a version, its hash or its size. | WF-04 | Đinh Gia Huy |
| F-57 | Sample folders and signing file versions | Head of department | Main: Creating a sample's folders adds its raw-data and result folders. Rule: A signature belongs to one version and its hash; a newer version shows as not signed. Exception: Creating the folders twice adds no second folder; rejecting a version without a reason is refused. | WF-04 | Đinh Gia Huy |

## 4. Non-Functional Requirements

| ID | Quality attribute | Metric | Threshold | Measurement method | Owner |
|---|---|---|---|---|---|
| NFR-01 | Performance | 95th percentile response time of common operations | ≤ 3 s with 30 concurrent users | Load test with k6 on staging | Lê Sỹ Tuyền |
| NFR-02 | Availability | Time until the system answers again after one site is lost | ≤ 60 s | Failover drill on staging, checking the health address every 5 s | Lê Sỹ Tuyền |
| NFR-03 | Recoverability | Data lost, and time to restore from backups | ≤ 5 min; ≤ 1 h | Restore drill into a new database cluster | Lê Sỹ Tuyền |
| NFR-04 | Security | Protected actions that succeed without permission; secrets found in the repository | 0; 0 | Permission tests per role in CI; GitGuardian scan on every pull request | Phạm Anh Kiệt |
| NFR-05 | Auditability | Signatures, approvals or deletions missing from the audit report | 0 | Audit tests in CI | Lê Nguyễn Hoàng Anh |
| NFR-06 | Maintainability | Pull requests merged with a failing check: SonarQube Cloud quality gate, branch and title conventions, linting, type checking, unit and browser tests | 0 | GitHub requires every check to pass before a merge to main | Lê Sỹ Tuyền |

## 5. Out of Scope

| Item | Reason |
|---|---|
| Invoices through an e-invoice provider | Invoices are Odoo invoices; a certified provider contract is outside the project. |
| Several laboratories in one installation | Each installation serves one laboratory company. |
| Customer price lists | Special prices are given as quotation discounts. |
| Odoo Enterprise apps | Every module installs on free Odoo Community. |

## 6. Workflow Commitments

### 6.1 Workflow WF-01

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|---|---|
| **Workflow name** | WF-01 – Order to confirmed test request |
| **Owner (one member)** | Lê Sỹ Tuyền |
| **Supporting members** | Phạm Anh Kiệt (PayOS online payment F-09, permissions) |
| **Actors** | Customer, Sales, Head of Sales, Accountant |
| **Trigger** | A customer submits a cart on the portal, or sales creates an order for a customer |
| **Business outcome** | Order confirmed; test request paid with its expected samples |
| **Main entity state path** | Quotation draft → sent → accepted, approved by the Head of Sales before it is sent when above the thresholds; order submitted → claimed → confirmed; test request draft → confirmed → paid |
| **Activity diagram (Report 4 section)** | Report 4 / docs/diagrams/business-flows.md §WF-01 |

**Features in this workflow**

| Feature ID | Feature | Acceptance criteria | Test case IDs |
|---|---|---|---|
| F-04 | Portal cart, drafts and reorder | Main: A visitor fills a cart without signing in and submits it with their details. Rule: A package is ordered whole; reordering creates a draft at current prices and leaves the original unchanged. Exception: Submitting without an e-mail address is refused. | TC-F04-01..03 |
| F-05 | Order claiming and orders on behalf | Main: A salesperson claims a submitted order, or creates one for a customer who called. Rule: A salesperson sees only their orders and the unclaimed queue; the Head of Sales sees all. Exception: An order already claimed cannot be claimed by another salesperson. | TC-F05-01..03 |
| F-06 | Quotation and revisions | Main: Sales confirms the ways and the expected samples calculated from the sampling requirements, applies a percentage discount and sends the quotation. Rule: Each item copies its price, discount and tax rate, so a sent quotation never changes; a revision makes the old version read-only. Exception: A quotation without tests cannot be sent, nor one above the Head of Sales's thresholds before it is approved. | TC-F06-01..03 |
| F-09 | Advance invoice and online payment | Main: Sending a quotation with an advance creates exactly one posted advance invoice; paying it accepts the quotation. Rule: Online payment accepts any amount up to the remaining balance; a bank transfer or cash payment counts once the accountant confirms it. Exception: A repeated PayOS notification records no second payment. | TC-F09-01..03 |
| F-10 | Quotation approval and order confirmation | Main: A quotation above the Head of Sales's thresholds is approved before it is sent; confirming the accepted order creates its test request with the calculated expected samples. Rule: Quotations within the thresholds and confirmed orders need no approval; the customer pays the balance before testing unless they may order on credit. Exception: The salesperson who requested approval cannot approve; a declined quotation goes back to Sales with the reason; a quotation awaiting approval cannot be edited. | TC-F10-01..03 |

**Demo scenarios**

| Scenario ID | Type | Preconditions and data | Main steps | Expected result | Presenter |
|---|---|---|---|---|---|
| DS-01a | Main | Demo customer; 2 parameters in the catalog; PayOS sandbox | Submit cart → sales claims the order, confirms the ways and 1 expected sample, applies a discount above the threshold and requests approval → Head of Sales signs → sales sends with full advance → customer pays → sales confirms the order | Order confirmed; test request paid with 1 expected sample | Lê Sỹ Tuyền |
| DS-01b | Exception · WF-01-E1 | Sent quotation | Customer declines with a reason → sales revises and sends the new version → customer pays | Old version read-only; order goes on with the revision | Lê Sỹ Tuyền |
| DS-01c | Exception · WF-01-E2 | Quotation past its payment deadline | Sales extends the deadline → customer pays | Expired flag cleared; quotation accepted | Lê Sỹ Tuyền |
| DS-01d | Exception · WF-01-E3 | Quotation with a discount above the threshold waiting for approval | Head of Sales rejects with a reason → sales adjusts the discount and requests again → Head of Sales signs | Sales told the reason; quotation approved, then sent | Lê Sỹ Tuyền |

| Scenario ID | Type | Preconditions and data | Main steps | Expected result | Presenter |
|---|---|---|---|---|---|
| DS-01e | Alternative · WF-01-A4 | Quotation sent with full advance | Customer pays by bank transfer → sales records the payment → accountant confirms it → quotation accepted → sales confirms the order | Payment counted once confirmed; order confirmed and test request paid | Lê Sỹ Tuyền |

**Committed test cases**

| Test type | Count | Automated in CI | Responsible member(s) |
|---|---|---|---|
| Feature tests (≥ 3 per feature) | 15 | 15 | Lê Sỹ Tuyền, Phạm Anh Kiệt |
| End-to-end workflow test (≥ 1) | 1 | 1 | Lê Sỹ Tuyền |
| State transition & NFR tests (18 state + 4 NFR: NFR-01..03, 06) | 22 | 19 | Lê Sỹ Tuyền |
| Total for this workflow | 38 | 35 | Lê Sỹ Tuyền |

**Pass criteria at Defense 1.1**

| Criterion | Committed |
|---|---|
| Runs end-to-end on staging with seeded data | Yes |
| Main scenario passes | Yes (100%) |
| Exception scenarios passing | 4 / 4 (100%; ≥ 80% threshold) |
| Test pass rate for this workflow | ≥ 95% overall; 100% main-path tests of mandatory features |
| Open Critical / High defects | 0 |

**Reviewer note**

| Note | |
|---|---|

## 6.2 Workflow WF-02

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|---|---|
| **Workflow name** | WF-02 – Sample reception and handover |
| **Owner (one member)** | Nguyễn Nam Hoàng |
| **Supporting members** | Đinh Gia Huy (test request lab codes F-15, sample states) |
| **Actors** | Customer, Sample delivery staff, Head of department |
| **Trigger** | Samples arrive at the laboratory, or sample delivery staff collect them from the customer |
| **Business outcome** | Samples coded, labelled and received; tasks queued in each department |
| **Main entity state path** | Sample expected → received, or → on hold → received, or → rejected, decided by its condition before receipt; test request paid → sample received |
| **Activity diagram (Report 4 section)** | Report 4 / docs/diagrams/business-flows.md §WF-02 |

**Features in this workflow**

| Feature ID | Feature | Acceptance criteria | Test case IDs |
|---|---|---|---|
| F-15 | Test requests and lab codes | Main: Staff create a test request with its samples, parameters, expected date and report delivery. Rule: Each sample gets a unique lab code, numbered again each year; a request staff create skips the paid step. Exception: Confirming without a sample, or a sample without a parameter, is refused; testers see only the lab code. | TC-F15-01..03 |
| F-16 | Sample collection and handover record | Main: Staff claim the collection task and plan its time; it appears on their schedule and to-do list, with a reminder. Rule: The customer confirms the actual samples and signs the handover record before labels are printed; signing stores the signature, time and device and e-mails the PDF. Exception: A signed handover record cannot be edited; a correction is a new record. | TC-F16-01..03 |
| F-17 | Reception, labels and department handover | Main: The condition is checked first; a sample in good condition is recorded as received with its time, amount and storage, and one testing task per sample test is queued. Rule: The label shows the lab code, request code, due date and QR code, never the customer. Exception: A label scanned by a person outside the department is refused. | TC-F17-01..03 |
| F-18 | Insufficient and rejected samples | Main: An insufficient sample goes on hold before receipt and its head of department is told. Rule: Asking for more creates a collection task recording who asked and when; more sample returns it to received. Exception: An unusable sample is rejected by the system with its condition note and gets no testing task. | TC-F18-01..03 |

**Demo scenarios**

| Scenario ID | Type | Preconditions and data | Main steps | Expected result | Presenter |
|---|---|---|---|---|---|
| DS-02a | Main | Paid request with 1 expected sample | Customer hands over the sample → staff record receipt in good condition → customer confirms and signs the handover record → signed PDF e-mailed → print label → department scans label | Sample received; task queued; signed PDF e-mailed | Nguyễn Nam Hoàng |
| DS-02b | Exception · WF-02-E1 | Expected sample arrives insufficient | Record it insufficient → head asks for more → staff collect through the collection task → record receipt | Sample on hold, then received; tasks queued | Nguyễn Nam Hoàng |
| DS-02c | Exception · WF-02 unusable sample | Request with two samples, one unusable | Record the unusable one as damaged with a condition note → system rejects it | That sample rejected with its note; the other goes on to testing | Nguyễn Nam Hoàng |
| DS-02d | Exception · WF-02-E2 | Signed handover record with a wrong quantity | Staff make a corrected record → customer signs it | New record signed; the old one kept unchanged | Nguyễn Nam Hoàng |

**Committed test cases**

| Test type | Count | Automated in CI | Responsible member(s) |
|---|---|---|---|
| Feature tests (≥ 3 per feature) | 12 | 12 | Nguyễn Nam Hoàng, Đinh Gia Huy |
| End-to-end workflow test (≥ 1) | 1 | 1 | Nguyễn Nam Hoàng |
| State transition tests (≥ 1 per transition) | 11 | 11 | Nguyễn Nam Hoàng |
| Total for this workflow | 24 | 24 | Nguyễn Nam Hoàng |

**Pass criteria at Defense 1.1**

| Criterion | Committed |
|---|---|
| Runs end-to-end on staging with seeded data | Yes |
| Main scenario passes | Yes (100%) |
| Exception scenarios passing | 3 / 3 (100%; ≥ 80% threshold) |
| Test pass rate for this workflow | ≥ 95% overall; 100% main-path tests of mandatory features |
| Open Critical / High defects | 0 |

**Reviewer note**

| Note | |
|---|---|

## 6.3 Workflow WF-03

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|---|---|
| **Workflow name** | WF-03 – Testing assignment and scheduling |
| **Owner (one member)** | Phạm Anh Kiệt |
| **Supporting members** | Đinh Gia Huy (machine scheduler D4), Lê Sỹ Tuyền (tasks and reminders) |
| **Actors** | Head of department, Tester |
| **Trigger** | A testing task enters the department queue |
| **Business outcome** | Test assigned and booked in its agreed way on the fit machine that finishes it soonest before its due date, after the head approves the booking |
| **Main entity state path** | Sample test waiting → assigned → testing; booking proposed → scheduled → in progress → done |
| **Activity diagram (Report 4 section)** | Report 4 / docs/diagrams/business-flows.md §WF-03 |

**Features in this workflow**

| Feature ID | Feature | Acceptance criteria | Test case IDs |
|---|---|---|---|
| F-23 | Task assignment and claiming | Main: Assigning a sample assigns all its testing tasks in the department to one person, who is notified. Rule: An assigned task leaves the queue and appears in the assignee's to-do list. Exception: A claimed task cannot be claimed again; another department's task cannot be claimed; reassigning without a reason is refused. | TC-F23-01..03 |
| F-24 | Booking proposal and head approval | Main: The system proposes a booking of the agreed way on the fit machine that finishes soonest before the due date; when it is feasible, the head approves it, the tester is notified, and the tester, required locations and machine are reserved together. Rule: A booking takes the run time rounded up to 15-minute slots; urgent tests first, then first in first out. Exception: Bookings of one machine never overlap; with no feasible booking the head reviews other dates, planned OverTime or outsourcing. | TC-F24-01..03 |
| F-25 | Schedules and reminders | Main: A booking appears on the person's calendar, the machine's schedule and the to-do list. Rule: A reminder | TC-F25-01..03 |

| Feature ID | Feature | Acceptance criteria | Test case IDs |
|---|---|---|---|
| | | arrives 15 minutes before it starts, once on each channel. Exception: An event the administrator turned off notifies nobody; a mandatory event cannot be turned off. | |

**Demo scenarios**

| Scenario ID | Type | Preconditions and data | Main steps | Expected result | Presenter |
|---|---|---|---|---|---|
| DS-03a | Main | Received sample; two fit machines with different free slots | Head assigns the sample → tester is notified and views the proposed booking → head approves it → tester is notified → tester, locations and machine reserved | Booking on the machine that finishes first, before the due date, on the tester's schedule | Phạm Anh Kiệt |
| DS-03b | Exception · WF-03-E1 | Task assigned to a tester who is absent | Head moves it to another tester with a reason → new tester notified → head approves the proposed booking | Reason kept on the task; test booked in time | Phạm Anh Kiệt |
| DS-03c | Exception · WF-03-E2 | The only machine for the test is overdue for calibration | System leaves out the overdue machine and finds no feasible slot → suggests outsourcing → head accepts it | No booking; test outsourced | Phạm Anh Kiệt |

**Committed test cases**

| Test type | Count | Automated in CI | Responsible member(s) |
|---|---|---|---|
| Feature tests (≥ 3 per feature) | 9 | 9 | Phạm Anh Kiệt, Đinh Gia Huy |
| End-to-end workflow test (≥ 1) | 1 | 1 | Phạm Anh Kiệt |
| State transition & NFR tests (10 state + 1 NFR: NFR-04) | 11 | 11 | Phạm Anh Kiệt |
| Total for this workflow | 21 | 21 | Phạm Anh Kiệt |

**Pass criteria at Defense 1.1**

| Criterion | Committed |
|---|---|
| Runs end-to-end on staging with seeded data | Yes |
| Main scenario passes | Yes (100%) |
| Exception scenarios passing | 2 / 2 (100%; ≥ 80% threshold) |
| Test pass rate for this workflow | ≥ 95% overall; 100% main-path tests of mandatory features |
| Open Critical / High defects | 0 |

**Reviewer note**

| Note | |
|---|---|

## 6.4 Workflow WF-04

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|---|---|
| **Workflow name** | WF-04 – Result entry and approval |
| **Owner (one member)** | Đinh Gia Huy |
| **Supporting members** | Phạm Anh Kiệt (laboratory folders, file versions and sync F-53, F-54, F-57) |
| **Actors** | Tester, Head of department |
| **Trigger** | A booked test starts; its machine is marked busy until the testing task closes |
| **Business outcome** | Result approved against its limit; the result file kept as a signed version; sample completed |
| **Main entity state path** | Sample test testing → result entered → approved (→ testing when rejected); sample testing → awaiting approval → completed; result file version current or conflict, then signed; machine available → busy → available |
| **Activity diagram (Report 4 section)** | Report 4 / docs/diagrams/business-flows.md §WF-04, §SF-11 |

**Features in this workflow**

| Feature ID | Feature | Acceptance criteria | Test case IDs |
|---|---|---|---|
| F-30 | Testing method steps | Main: The tester sees the method's steps with number, duration, what to do and note, in number order. Rule: Two steps of one method cannot share a number. Exception: A tester cannot change a step. | TC-F30-01..03 |
| F-32 | Regulations and limits | Main: When a regulation comparison is requested, the result shows the related limit copied with its unit and is assessed. Rule: Later changes to the regulation leave existing results unchanged; a regulation holds one limit per parameter. Exception: A result with a requested comparison cannot be sent without its assessment. | TC-F32-01..03 |

| Feature ID | Feature | Acceptance criteria | Test case IDs |
|---|---|---|---|
| F-34 | Result entry and approval | Main: The tester enters, signs and sends a result, which closes the testing task; the head approves it. Rule: A rejected result is kept, and a retest adds a new result with a new booking; only the latest is sent for approval. Exception: A tester cannot approve their own result, but the head may approve an entry they made; a rejection opens a new testing task and tells the tester the reason. | TC-F34-01..03 |
| F-35 | Unusual result warning | Main: A numeric result far from approved results of the same parameter, way of testing and sample type warns before sending. Rule: A result near the regulation limit warns; with too few approved results only the limit check runs. Exception: Sending anyway keeps the warning, which the approving head sees; the warning never names a customer. | TC-F35-01..03 |
| F-36 | Sample test cancellation | Main: The head cancels a waiting, assigned or testing sample test with a reason; its task and bookings are cancelled. Rule: A sample whose remaining tests are approved completes; a cancelled test is left out of the report. Exception: Cancelling without a reason, or a test with an approved result, is refused. | TC-F36-01..03 |
| F-53 | Laboratory folders and sync | Main: Saving a file in a folder bound in Medilab Sync adds a version with the tester, the computer, the size and the hash. Rule: Only files matching the folder's sync rules are uploaded; content the file already has is not stored twice. Exception: A person who is not a member of the folder, or cannot see its sample, gets nothing from it. | TC-F53-01..03 |
| F-54 | File versions, conflicts and restore | Main: Restoring version 3 of a file with 5 versions adds version 6 with the content of version 3. Rule: A file uploaded from an older version is kept as a conflict version and the current version stays. Exception: No one can change or delete a version, its hash or its size. | TC-F54-01..03 |
| F-57 | Sample folders and signing file versions | Main: Creating a sample's folders adds its raw-data and result folders. Rule: A signature belongs to one version and its hash; a newer version shows as not signed. Exception: Creating the folders twice adds no second folder; rejecting a version without a reason is refused. | TC-F57-01..03 |

**Demo scenarios**

| Scenario ID | Type | Preconditions and data | Main steps | Expected result | Presenter |
|---|---|---|---|---|---|
| DS-04a | Main | Booked test on a sample with a regulation; the sample's result folder bound on the tester's computer | Booking starts and the machine is marked busy → tester saves the instrument's result file → Medilab Sync uploads it → head signs the file version → tester enters the result and its assessment → sends and signs → head approves the result | Result keeps the copied limit; file version signed; sample completed | Đinh Gia Huy |
| DS-04b | Exception · WF-04-E1 | Result sent for approval | Head rejects with a reason → tester retests with a new booking → head approves the new result | Rejected result kept on record; new result approved | Đinh Gia Huy |
| DS-04c | Exception · WF-04-E2 | Result far from past approved results | Tester sees the warning → sends anyway → head approves | Warning kept on the result and shown to the head | Đinh Gia Huy |
| DS-04d | Exception · WF-04-E3 | Test that cannot be done on the sample | Head cancels it with a reason | Sample completes with its other tests; cancelled test left out of the report | Đinh Gia Huy |
| DS-04e | Exception · WF-04-E4 | Same result file changed on two computers from one version | Second tester's upload becomes a conflict → they take the server's version → save their changes again | Both versions kept; local file backed up; changes saved again | Đinh Gia Huy |
| DS-04f | Exception · WF-04-E5 | Result file version waiting for signature | Head rejects it with a reason → tester saves a corrected version → head signs | Corrected version signed | Đinh Gia Huy |
| DS-04g | Exception · WF-04-E6 | Online editing service down | Tester saves the result file in its own program | Medilab Sync uploads it; history and restore keep working | Đinh Gia Huy |

**Committed test cases**

| Test type | Count | Automated in CI | Responsible member(s) |
|---|---|---|---|
| Feature tests (≥ 3 per feature) | 24 | 24 | Đinh Gia Huy, Phạm Anh Kiệt |
| End-to-end workflow test (≥ 1) | 1 | 1 | Đinh Gia Huy |
| State transition tests (≥ 1 per transition) | 16 | 16 | Đinh Gia Huy, Phạm Anh Kiệt |
| Total for this workflow | 41 | 41 | Đinh Gia Huy |

**Pass criteria at Defense 1.1**

| Criterion | Committed |
|---|---|
| Runs end-to-end on staging with seeded data | Yes |
| Main scenario passes | Yes (100%) |
| Exception scenarios passing | 6 / 6 (100%; ≥ 80% threshold) |
| Test pass rate for this workflow | ≥ 95% overall; 100% main-path tests of mandatory features |
| Open Critical / High defects | 0 |

**Reviewer note**

| Note | |
|---|---|

## 6.5 Workflow WF-05

| | |
|---|---|
| **Workflow name** | WF-05 – Outsourced testing |
| **Owner (one member)** | Lê Nguyễn Hoàng Anh |
| **Supporting members** | Đinh Gia Huy (outsourcing suggestions from scheduler) |
| **Actors** | Head of department, Sample delivery staff, Subcontractor |
| **Trigger** | No machine can finish a test before its due date, or the head outsources a test urgently |
| **Business outcome** | Subcontractor's result approved |
| **Main entity state path** | Sample test waiting → outsourced → result entered → approved; dispatch draft → sent → received, or draft → cancelled before any result |
| **Activity diagram (Report 4 section)** | Report 4 / docs/diagrams/business-flows.md §WF-05 |

**Features in this workflow**

| Feature ID | Feature | Acceptance criteria | Test case IDs |
|---|---|---|---|
| F-38 | Outsourcing suggestions and dispatches | Main: Accepting an outsourcing suggestion, or outsourcing a test urgently with a reason, puts its sample tests in a dispatch; when staff carry it, the system records the date sent, who took it and the expected date. Rule: Only a subcontracted way with the test's parameter and method can be chosen; the customer is neither told nor asked. Exception: A dispatch whose tests have results cannot be cancelled; a failed delivery goes back to sample reception for the same sample. | TC-F38-01..03 |
| F-40 | Subcontractor results | Main: The subcontractor receives the samples and enters the results of the tests sent to it, and the head is notified. Rule: The result shows that user as its author and goes straight to the head for approval like any other. Exception: A subcontractor user sees only its own dispatches and never a sample's customer. | TC-F40-01..03 |

**Demo scenarios**

| Scenario ID | Type | Preconditions and data | Main steps | Expected result | Presenter |
|---|---|---|---|---|---|
| DS-05a | Main | Only machine for the test is under repair; accredited subcontractor seeded | System suggests outsourcing → head accepts it → staff carry the dispatch and the system records it → subcontractor receives the samples and enters the result → head approves | Dispatch received; sample test approved | Lê Nguyễn Hoàng Anh |
| DS-05b | Exception · WF-05-E1 | Subcontractor's result waiting for approval | Head rejects it → subcontractor enters a new result → head approves | Test back to outsourced, then approved | Lê Nguyễn Hoàng Anh |
| DS-05c | Exception · WF-05-E2 | Draft dispatch that cannot go out | Head cancels it → puts its tests in another dispatch | Tests stay outsourced and go with the new dispatch | Lê Nguyễn Hoàng Anh |
| DS-05d | Alternative · WF-05-A3 | Machine breaks; test assigned with no outsourcing suggestion | Head requests outsourcing with a reason → puts the test in a dispatch → staff carry it to the subcontractor | Reason shown on the test; dispatch on its way to the subcontractor | Lê Nguyễn Hoàng Anh |

**Committed test cases**

| Test type | Count | Automated in CI | Responsible member(s) |
|---|---|---|---|
| Feature tests (≥ 3 per feature) | 6 | 6 | Lê Nguyễn Hoàng Anh |
| End-to-end workflow test (≥ 1) | 1 | 1 | Lê Nguyễn Hoàng Anh |
| State transition & NFR tests (10 state + 1 NFR: NFR-05) | 11 | 11 | Lê Nguyễn Hoàng Anh |
| Total for this workflow | 18 | 18 | Lê Nguyễn Hoàng Anh |

**Pass criteria at Defense 1.1**

| Criterion | Committed |
|---|---|
| Runs end-to-end on staging with seeded data | Yes |
| Main scenario passes | Yes (100%) |
| Exception scenarios passing | 3 / 3 (100%; ≥ 80% threshold) |
| Test pass rate for this workflow | ≥ 95% overall; 100% main-path tests of mandatory features |
| Open Critical / High defects | 0 |

**Reviewer note**

| Note | |
|---|---|

## 6.6 Workflow WF-06

| | |
|---|---|
| **Workflow name** | WF-06 – Report signing, release and delivery |
| **Owner (one member)** | Phạm Anh Kiệt |
| **Supporting members** | Đinh Gia Huy (test report generation and signing chain) |
| **Actors** | Lab head, Head of department, Sample delivery staff, Customer |
| **Trigger** | Every sample test of a request is approved or cancelled |
| **Business outcome** | Signed report issued, delivered and downloadable on the portal |
| **Main entity state path** | Test report draft → signing → issued (→ replaced); test request completed → reported |
| **Activity diagram (Report 4 section)** | Report 4 / docs/diagrams/business-flows.md §WF-06 |

**Features in this workflow**

| Feature ID | Feature | Acceptance criteria | Test case IDs |
|---|---|---|---|
| F-43 | Report signing and digital signature | Main: When every test of the request is approved or cancelled, the system creates the read-only draft report, validates its data, reserves a number and prepares the PDF in the request's language; the lab head reviews it, the PDF is digitally signed and stored, and issue makes the number official. Rule: The report prints the approving heads and the lab head, not the testers, and leaves out cancelled tests. Exception: With Viettel Sign down nothing is marked signed and signing is retried; rejecting without a reason or a named result is refused. | TC-F43-01..03 |
| F-45 | Report delivery and download | Main: Staff record the delivery of an issued report; an e-mail delivery sends the PDF. Rule: An issued report waits in the delivery list until its delivery is recorded. Exception: An unissued report cannot be delivered; a customer downloads only their own reports. | TC-F45-01..03 |
| F-47 | Notifications and preferences | Main: An issued report notifies the customer on the channels they chose. Rule: A new customer is notified by e-mail without configuring anything; a changed preference applies to the next notification. Exception: A mandatory notification cannot be turned off. | TC-F47-01..03 |

**Demo scenarios**

| Scenario ID | Type | Preconditions and data | Main steps | Expected result | Presenter |
|---|---|---|---|---|---|
| DS-06a | Main | Request with all results approved; Viettel Sign sandbox | System prepares the PDF → lab head reviews and approves it → PDF digitally signed and stored → report issued → staff records e-mail delivery → customer downloads | Report issued with number and signed PDF; request reported | Phạm Anh Kiệt |
| DS-06b | Exception · WF-06-E2 | Draft report | Lab head rejects naming a result → head of department approves it again → lab head signs | Result back in the department queue, then the report issued | Phạm Anh Kiệt |
| DS-06c | Exception · WF-06-E1 | Viettel Sign unreachable | Lab head approves → signing retried once Viettel Sign answers | Report not marked signed until the retry succeeds, then issued | Phạm Anh Kiệt |

**Committed test cases**

| Test type | Count | Automated in CI | Responsible member(s) |
|---|---|---|---|
| Feature tests (≥ 3 per feature) | 9 | 9 | Phạm Anh Kiệt, Đinh Gia Huy |
| End-to-end workflow test (≥ 1) | 1 | 1 | Phạm Anh Kiệt |
| State transition tests (≥ 1 per transition) | 9 | 9 | Phạm Anh Kiệt |
| Total for this workflow | 19 | 19 | Phạm Anh Kiệt |

**Pass criteria at Defense 1.1**

| Criterion | Committed |
|---|---|
| Runs end-to-end on staging with seeded data | Yes |
| Main scenario passes | Yes (100%) |
| Exception scenarios passing | 2 / 2 (100%; ≥ 80% threshold) |
| Test pass rate for this workflow | ≥ 95% overall; 100% main-path tests of mandatory features |
| Open Critical / High defects | 0 |

### Reviewer note

| Note | |
|---|---|

## 7. Individual Commitments

### 7.1 Member 1

| | |
|---|---|
| **Full name** | Nguyễn Nam Hoàng |
| **Student ID** | SE183190 |
| **Role in team** | Team Leader & Developer |

**Ownership**

| Type | Items (IDs / names) | Notes |
|---|---|---|
| Main workflow(s) owned | WF-02 (Sample reception and handover) | sole owner |
| Mandatory features owned | F-15, F-16, F-17, F-18 (4 features) | sample reception, lab codes, QR labels, handover records |
| Modules / entities owned | sol_laboratory (SampleHandover, SampleReception, SampleCondition), sol_commerce (Customer Portal & Catalog) | reception & portal UI |
| Design decisions owned (D-IDs) | D1 (Platform and modules – Odoo 20 Community) | architecture & module boundaries |
| NFRs responsible | Supports NFR-01, NFR-04 on reception and portal | portal and reception verification |

**Delivery commitments to Defense 1.1**

| Commitment | Committed (count / IDs) | Due |
|---|---|---|
| Functional test cases written | 24 (TC-F15-01..03 to TC-F18-01..03, 1 E2E, 11 state transitions) | 12/11/2026 |
| Of which automated in CI | 24 (100%) | 12/11/2026 |
| Demo scenarios presented | 4 (DS-02a Main; DS-02b, DS-02c, DS-02d Exception) | Defense 1.1 |
| Required fixes owned (section 9) | Coordinates team verification of RF-01..RF-10 before Defense 1.1 | 12/11/2026 |
| Documentation sections (Report / section) | Report 1 (Project Overview), Report 3 (Reception & Portal SRS), BM01/BM03/BM04 coordination | 05/11/2026 |

**Ownership verification at Review 2 (completed by reviewers)**

| Result | Weak points observed | Follow-up (by whom, by when) |
|---|---|---|
| ☐ *Verified*<br>☐ *Partly verified*<br>☐ *Not verified* | | |

**Member confirmation**

| | |
|---|---|
| **Signature / date** | Nguyễn Nam Hoàng – |

### 7.2 Member 2

| | |
|---|---|
| **Full name** | Lê Sỹ Tuyền |
| **Student ID** | SE190797 |
| **Role in team** | DevOps & Backend Developer |

**Ownership**

| Type | Items (IDs / names) | Notes |
|---|---|---|
| Main workflow(s) owned | WF-01 (Order to confirmed test request) | sole owner |
| Mandatory features owned | F-04, F-05, F-06, F-09, F-10 (5 features) | portal cart, order claiming, quotation revisions, quotation approval |
| Modules / entities owned | sol_commerce (SaleOrder, QuotationVersion, OrderApproval, CancellationRequest); sol_laboratory (Schedule, TodoItem); Kubernetes & GitOps infra | orders, schedules & k3s/CNPG/Argo CD |
| Design decisions owned (D-IDs) | D5 (Deployment – Dual-site k3s, CloudNativePG, S3 backups, Argo CD) | high availability & CI/CD |
| NFRs responsible | NFR-01 (Performance), NFR-02 (Availability), NFR-03 (Recoverability), NFR-06 (Maintainability) | k6 load test, failover & restore drills, CI gates |

**Delivery commitments to Defense 1.1**

| Commitment | Committed (count / IDs) | Due |
|---|---|---|
| Functional test cases written | 38 (TC-F04-01..03 to TC-F10-01..03, 1 E2E, 18 state transitions, 4 NFR tests) | 12/11/2026 |

| Commitment | Committed (count / IDs) | Due |
|---|---|---|
| Of which automated in CI | 35 (35 in GitHub Actions CI + 3 staging drills for NFR-01..03) | 12/11/2026 |
| Demo scenarios presented | 5 (DS-01a Main; DS-01b, DS-01c, DS-01d Exception; DS-01e Alternative) | Defense 1.1 |
| Required fixes owned (section 9) | RF-03 (SR-07 Location model), RF-04 (SR-04 NFR table), RF-05 (SR-05), RF-07 (SR-10), RF-08 (SR-14), RF-10 (SR-13) | 12/11/2026 |
| Documentation sections (Report / section) | Report 3 (E-commerce SRS & NFRs), Report 4 (Architecture, Deployment & ERD diagrams) | 05/11/2026 |

**Ownership verification at Review 2 (completed by reviewers)**

| Result | Weak points observed | Follow-up (by whom, by when) |
|---|---|---|
| ☐ *Verified*<br>☐ *Partly verified*<br>☐ *Not verified* | | |

**Member confirmation**

| Signature / date | Lê Sỹ Tuyền – |
|---|---|

## 7.3 Member 3

| Full name | Đinh Gia Huy |
|---|---|
| Student ID | SE193759 |
| Role in team | Lead Backend Developer |

**Ownership**

| Type | Items (IDs / names) | Notes |
|---|---|---|
| Main workflow(s) owned | WF-04 (Result entry and approval) | sole owner |
| Mandatory features owned | F-30, F-32, F-34, F-35, F-36, F-53, F-54, F-57 (8 features in WF-04; co-delivers F-53, F-54, F-57 with Phạm Anh Kiệt) | methods, regulation limits, results, retests, cancellations |
| Modules / entities owned | sol_laboratory (TestRequest, Sample, SampleTest, TestResult, Regulation, TestingMethod, MachineBooking) | core laboratory domain & scheduler |
| Design decisions owned (D-IDs) | D3 (Approving is signing), D4 (Machine scheduling in 15-minute slots) | signing chains & machine queue |
| NFRs responsible | Co-owns NFR-04 (Sample anonymisation R2) and NFR-05 (Result & signature auditability) | role & anonymisation tests |

**Delivery commitments to Defense 1.1**

| Commitment | Committed (count / IDs) | Due |
|---|---|---|
| Functional test cases written | 41 (TC-F30..F36, TC-F53, TC-F54, TC-F57, 1 E2E, 16 state transitions) | 12/11/2026 |
| Of which automated in CI | 41 (100%) | 12/11/2026 |
| Demo scenarios presented | 7 (DS-04a Main; DS-04b, DS-04c, DS-04d, DS-04e, DS-04f, DS-04g Exception) | Defense 1.1 |
| Required fixes owned (section 9) | RF-01 (SR-11 implement F-35 unusual result warning), RF-10 (SR-13 Iteration 2 scheduling extensions) | 05/11/2026 |
| Documentation sections (Report / section) | Report 3 (Laboratory SRS), Report 4 (State machines, signing chain & scheduling algorithms) | 05/11/2026 |

**Ownership verification at Review 2 (completed by reviewers)**

| Result | Weak points observed | Follow-up (by whom, by when) |
|---|---|---|
| ☐ *Verified*<br>☐ *Partly verified*<br>☐ *Not verified* | | |

**Member confirmation**

| Signature / date | Đinh Gia Huy – |
|---|---|

## 7.4 Member 4

| Full name | Phạm Anh Kiệt |
|---|---|
| Student ID | SE190051 |
| Role in team | Full-Stack & Systems Developer |

**Ownership**

| Type | Items (IDs / names) | Notes |
|---|---|---|
| Main workflow(s) owned | WF-03 (Testing assignment and scheduling), WF-06 (Report signing, release and delivery) | owns 2 main workflows |
| Mandatory features owned | F-23, F-24, F-25 (WF-03) and F-43, F-45, F-47 (WF-06) (6 workflow features + laboratory file sync infrastructure F-53, F-54, F-57) | assignment, scheduling, digital signing, delivery, notifications, Medilab Sync |
| Modules / entities owned | sol_laboratory (Role/Permission, LaboratoryFolder, LaboratoryFile, LaboratoryUpload, TestReport, Notification); sol_commerce (PayOS); medilab-sync (Rust/Tauri) | permissions, S3 sync, Viettel Sign, PayOS, notifications |
| Design decisions owned (D-IDs) | D2 (Role & permission matrix generating Odoo groups and record rules) | access control across UI & API |
| NFRs responsible | NFR-04 (Security – 0 unauthorized actions across roles; 0 secrets in repo) | permission test suite & GitGuardian |

**Delivery commitments to Defense 1.1**

| Commitment | Committed (count / IDs) | Due |
|---|---|---|
| Functional test cases written | 40 (21 in WF-03 + 19 in WF-06, including NFR-04 permission tests) | 12/11/2026 |
| Of which automated in CI | 40 (100%) | 12/11/2026 |
| Demo scenarios presented | 6 (DS-03a, DS-06a Main; DS-03b, DS-03c, DS-06b, DS-06c Exception) | Defense 1.1 |
| Required fixes owned (section 9) | RF-02 (SR-12 migrate medilab-sync to /medilab/api/v1), RF-06 (SR-09 upload state diagram), RF-09 (SR-15 folder department signing) | 05/11/2026 |
| Documentation sections (Report / section) | Report 3 & Report 4 (Permissions spec, PayOS spec, Laboratory Files & Medilab Sync spec) | 05/11/2026 |

**Ownership verification at Review 2 (completed by reviewers)**

| Result | Weak points observed | Follow-up (by whom, by when) |
|---|---|---|
| ☐ *Verified*<br>☐ *Partly verified*<br>☐ *Not verified* | | |

**Member confirmation**

| | |
|---|---|
| **Signature / date** | Phạm Anh Kiệt – |

## 7.5 Member 5

| | |
|---|---|
| **Full name** | Lê Nguyễn Hoàng Anh |
| **Student ID** | SE184416 |
| **Role in team** | Backend Developer |

**Ownership**

| Type | Items (IDs / names) | Notes |
|---|---|---|
| Main workflow(s) owned | WF-05 (Outsourced testing) | sole owner |
| Mandatory features owned | F-38, F-40 (2 features) | outsourcing suggestions, dispatches, subcontractor portal results |
| Modules / entities owned | sol_laboratory (Subcontractor, OutsourcingSuggestion, SubcontractDispatch, AuditLog); sol_inventory (Chemical, StockReceipt, StockIssue) | outsourcing, audit trail & chemical inventory |
| Design decisions owned (D-IDs) | Co-owns D3/SH-05 (Immutable audit log and subcontractor isolation) | audit trail & subcontractor access rules |
| NFRs responsible | NFR-05 (Auditability – 0 signatures, approvals or deletions missing from audit trail) | automated audit trail tests in CI |

**Delivery commitments to Defense 1.1**

| Commitment | Committed (count / IDs) | Due |
|---|---|---|
| Functional test cases written | 18 (TC-F38-01..03, TC-F40-01..03, 1 E2E, 10 state transitions, 1 NFR-05 test) | 12/11/2026 |
| Of which automated in CI | 18 (100%) | 12/11/2026 |
| Demo scenarios presented | 4 (DS-05a Main; DS-05b, DS-05c Exception; DS-05d Alternative) | Defense 1.1 |

| Commitment | Committed (count / IDs) | Due |
|---|---|---|
| Required fixes owned (section 9) | RF-03 (SR-07 align inventory.prisma with Iteration 2 scope), RF-10 (SR-13 direct outsourcing F-39) | 12/11/2026 |
| Documentation sections (Report / section) | Report 3 & Report 4 (Outsourced testing workflow WF-05, Audit trail SH-05, Inventory ERD) | 05/11/2026 |

**Ownership verification at Review 2 (completed by reviewers)**

| Result | Weak points observed | Follow-up (by whom, by when) |
|---|---|---|
| ☐ *Verified*<br>☐ *Partly verified*<br>☐ *Not verified* | | |

**Member confirmation**

| | |
|---|---|
| **Signature / date** | Lê Nguyễn Hoàng Anh – |

## 8. Summary and Cross-check

### 8.1 By workflow

| Workflow | Owner | Features | Demo scenarios (main / exception) | Test cases | Automated |
|---|---|---|---|---|---|
| WF-01 | Lê Sỹ Tuyền | 5 | 5 (1 / 4) | 38 | 35 |
| WF-02 | Nguyễn Nam Hoàng | 4 | 4 (1 / 3) | 24 | 24 |
| WF-03 | Phạm Anh Kiệt | 3 | 3 (1 / 2) | 21 | 21 |
| WF-04 | Đinh Gia Huy | 8 | 7 (1 / 6) | 41 | 41 |
| WF-05 | Lê Nguyễn Hoàng Anh | 2 | 4 (1 / 3) | 18 | 18 |
| WF-06 | Phạm Anh Kiệt | 3 | 3 (1 / 2) | 19 | 19 |
| Total | 5 members (6 workflows) | 25 | 26 (6 / 20) | 161 | 158 |

### 8.2 By member

| Member | Workflows owned | Features | Test cases | Automated | Demo scenarios | Review 2 verification |
|---|---|---|---|---|---|---|
| Nguyễn Nam Hoàng | WF-02 | 4 | 24 | 24 | 4 (1 / 3) | Pending Review 2 |
| Lê Sỹ Tuyền | WF-01 | 5 | 38 | 35 | 5 (1 / 4) | Pending Review 2 |
| Đinh Gia Huy | WF-04 | 8 | 41 | 41 | 7 (1 / 6) | Pending Review 2 |
| Phạm Anh Kiệt | WF-03, WF-06 | 6 | 40 | 40 | 6 (2 / 4) | Pending Review 2 |
| Lê Nguyễn Hoàng Anh | WF-05 | 2 | 18 | 18 | 4 (1 / 3) | Pending Review 2 |
| Total | 6 workflows | 25 | 161 | 158 | 26 (6 / 20) | — |

### 8.3 Committed Numbers

| | |
|---|---|
| **Number of main workflows** | 6 (WF-01 to WF-06) |
| **Number of demo scenarios (main / exception)** | 26 total (6 main / 20 exception) |
| **Committed functional test cases** | 161 (75 feature + 6 E2E workflow + 74 state transition + 6 NFR) |
| **Of which automated in CI** | 158 (98.1%; 3 staging drills for NFR-01..03 run manually) |

### 8.4 Eligibility Thresholds for the Viva

| Criterion | Minimum threshold | Agreed threshold |
|---|---|---|
| Main workflows | 100% run end-to-end | 100% (all 6 workflows WF-01..WF-06 run end-to-end on staging) |
| Demo scenarios | 100% main; ≥ 80% exception | 100% main (6/6); ≥ 80% exception (≥ 16/20 scenarios) |
| Test case count | ≥ committed number | ≥ 161 test cases |
| Test pass rate | ≥ 90% overall; 100% main-path tests of mandatory features | ≥ 95% overall; 100% main-path tests of all 25 mandatory features |
| Automation | ≥ 50% run in CI | 98.1% (158 / 161 test cases run automatically in GitHub Actions CI) |
| Defects | No open Critical or High defects | 0 open Critical or High defects |
| Individual | Each member demos the workflow they own and meets their delivery commitments | 100% of members demo their owned workflow(s) and pass ownership verification |

## 9. Required Fixes

| ID | Item | Related workflow | Owner | Due |
|---|---|---|---|---|
| RF-01 (SR-11) | Implement mandatory feature F-35 (Unusual result sanity-check warning in WF-04-A4) in sol_laboratory/models/test_result.py and automated tests TC-F35-01..03. | WF-04 | Đinh Gia Huy | 05/11/2026 |
| RF-02 (SR-12) | Migrate medilab-sync Rust desktop client (src-tauri/src/api.rs) from legacy /api/sync/v1/* to the /medilab/api/v1/* presigned S3 upload/download flow (#169, #170, API-14) and complete TC-F53-01..03. | WF-04 (SF-11) | Phạm Anh Kiệt | 05/11/2026 |
| RF-03 (SR-07) | Implement laboratory.location model (#117) in sol_laboratory and reconcile inventory.prisma entities (Preparation, StockTake, StockTakeLine, StockCharge, Budget, AccountingApiKey) with Iteration 2 scope. | WF-03 | Lê Sỹ Tuyền, Lê Nguyễn Hoàng Anh | 05/11/2026 |
| RF-04 (SR-04) | Add consolidated Non-Functional Requirements table (NFR-01..NFR-06 with metric, threshold and | All workflows | Lê Sỹ Tuyền | 29/10/2026 |

| ID | Item | Related workflow | Owner | Due |
|---|---|---|---|---|
| | measurement method) into docs/functional/shared.md and Report 3. | | | |
| RF-05 (SR-05) | Replace 'Not written yet' placeholder for Head of Sales in docs/functional/ecommerce_module/overview.md with summary of US-S13..US-S15. | WF-01 | Lê Sỹ Tuyền | 29/10/2026 |
| RF-06 (SR-09) | Update Laboratory File Upload state diagram in docs/functional/laboratory_module/overview.md to match laboratory_upload.py (pending → verifying → completed \| unchanged \| refused \| expired). | WF-04 (SF-11) | Phạm Anh Kiệt | 29/10/2026 |
| RF-07 (SR-10) | Update Order Cancellation Request state note in docs/diagrams/business-flows.md (SF-02) from 'open' to 'requested → accepted, declined or withdrawn'. | WF-01 (SF-02) | Lê Sỹ Tuyền | 29/10/2026 |
| RF-08 (SR-14) | Update docs/infrastructure/architecture.md and Report 4 diagram to remove unused EMQX broker and include the laboratory S3 bucket, ONLYOFFICE, PayOS, Viettel Sign, Brevo and FCM. | All workflows | Lê Sỹ Tuyền | 29/10/2026 |
| RF-09 (SR-15) | Link sample folders to sample test departments and enforce non-author department-head signing in laboratory_file_version_signing.py (#228). | WF-04 (SF-11) | Phạm Anh Kiệt | 05/11/2026 |
| RF-10 (SR-13) | Complete Iteration 2 non-mandatory alternative flows: SF-07 estimated result date (F-08), WF-03-A5 room booking (F-21, F-26), WF-03-A6/A7 urgent bumping (F-27), WF-03-A8 deadline/machine warning cron (F-28) and WF-05-A3 direct outsourcing (F-39). | WF-01, WF-03, WF-05 | Lê Sỹ Tuyền, Đinh Gia Huy, Lê Nguyễn Hoàng Anh | 12/11/2026 |

## 10. Review 2 Conclusion (completed by reviewers)

☐ Pass ☐ Conditional pass ☐ Re-review

## 11. Confirmation

| Role | Full name | Signature | Date |
|---|---|---|---|
| Reviewer 1 | | | |
| Reviewer 2 | | | |
| Supervisor | Nguyễn Thế Hoàng | | |
| Team leader | Nguyễn Nam Hoàng | | |